

Software Requirements Specification (SRS)

PhantomID — Universal Temporary Identity Layer

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Executive Summary

PhantomID is an ephemeral identity runtime layer that enables an **Initiator** to project their digital identity—including payment capabilities, digital credentials, health records, corporate authorizations, and digital signatures—onto an arbitrary foreign **Host Device** inside a cryptographically bounded, isolated execution bubble.

Upon task completion or termination triggers, the execution environment instantly destructs all session tokens, ephemeral encryption keys, cached records, and application states, ensuring **absolute zero-residue digital footprint**.

[!IMPORTANT] **Core Value Proposition: "Borrow. Do. Disappear."**

Unlike traditional cloud logins that persist cookies, tokens, and credentials on foreign hardware, PhantomID establishes a transient, verifiable cryptographic session that self-destructs upon lifecycle completion.

1. Introduction

1.1 Purpose

This System Requirements Specification (SRS) establishes the complete functional, non-functional, interface, security, and architectural specifications for **PhantomID**. It serves as the authoritative blueprint for engineering teams, security auditors, hackathon evaluation juries, and strategic enterprise partners.

1.2 Project Overview

Modern digital authentication rigidly binds a user's operational capabilities (UPI payment apps, corporate VPN credentials, DigiLocker storage) to a single physical device. When that hardware is incapacitated (battery exhaustion, physical damage, device seizure, travel loss), users face an unbridgeable dilemma: forfeit access or endure high-risk credentials leakage on foreign/borrowed devices.

PhantomID resolves this dilemma by decoupling identity from static physical hardware, replacing persistent device logins with **ephemeral, sandboxed micro-sessions** secured by Multi-Factor Biometric Attestation and zero-trust orchestration.

1.3 Scope

PhantomID governs the following operational domains:

- **Digital Financial Transactions:** Ephemeral UPI payments, merchant QR payments, on-demand balance inquiries.
- **Sovereign Document Access:** Read-only inspection of medical dossiers, government identification, and educational transcripts without local storage.
- **Corporate Identity Federation:** Temporary role-based access for contractors and employees on partner/client workstations.
- **Hospitality & Transit Authentication:** Touchless, self-expiring hotel check-ins, transit gates, and car rental verification.
- **Emergency Kiosk Access:** Disaster recovery and critical administrative transactions on untrusted public terminals.

1.4 Definitions & Acronyms

Term / Acronym	Definition
PhantomID	The overarching software system, client runtime, and cloud verification engine.
Session	A strictly time-limited, isolated, cryptographically signed micro-container execution state.
Host Device	Any smartphone, tablet, or terminal running PhantomID Host Runtime, owned by another party.
Initiator	The sovereign user initiating and authorizing the temporary session.
Passkey (FIDO2 / WebAuthn)	Asymmetric public-key authentication credential stored within a Secure Enclave.
MDM	Mobile Device Management — operating-system level control profile for enterprise hardware.
UPI	Unified Payments Interface — real-time payment network operated by NPCI.
Sandbox	An isolated in-memory runtime restricting filesystem, clipboard, and IPC mechanisms.
Trust Score	A dynamic numeric risk rating evaluated in real time prior to session creation.
DPDPA 2023	India's Digital Personal Data Protection Act, 2023.

1.5 References

- PhantomID Pitch Document v1.0 (*PhantomID_Pitch.pdf*)
- W3C Web Authentication: An API for accessing Public Key Credentials (WebAuthn Level 3)
- FIDO Alliance FIDO2 Specifications & CTAP2 Protocols
- OWASP Mobile Security Testing Guide (MSTG Level 2)
- National Payments Corporation of India (NPCI) UPI Procedural Guidelines
- India Digital Personal Data Protection Act (DPDPA), Gazette of India (2023)

2. Overall Description

2.1 Product Perspective & Market Positioning

PhantomID operates as an orthogonal access layer. It does not seek to displace payment gateways, digital banking applications, or sovereign document repositories. Instead, it functions as an **access conduit** that temporarily bridges

user identity into a foreign physical device.

[Architecture / Flow Diagram]

```
graph TD
    subgraph Persistent_Storage_Layer [Persistent Storage Layer]
        DL[DigiLocker / Gov Vault]
        Bank[Core Banking / UPI]
        Corp[Enterprise SSO / IAM]
    end

    subgraph PhantomID_Core [PhantomID Core]
        Auth[Zero-Trust Multi-Factor Engine]
        SessionMgr[Ephemeral Session Manager]
        Enclave[Encrypted Memory Capsule]
    end

    subgraph Host_Device [Host Device]
        HostApp[Host Device Runtime]
        Display[Zero-Residue Display Layer]
    end

    InitiatorDevice((Primary Phone / Passkey)) -->|Challenge Approval| Auth
    Auth --> SessionMgr
    Persistent_Storage_Layer -->|Encrypted Stream| Enclave
    Enclave -->|Restricted Render| Display
    Display -->|Auto-Purge on Exit| HostApp
```

Strategic Differentiation: Storage vs. Access

- **DigiLocker / Cloud Wallets:** Persistent document vaults. If accessed on a third-party device, browser history, session tokens, cached PDFs, and cookies persist indefinitely.
- **PhantomID:** Access mechanism only. It streams cryptographic proofs and temporary access tokens. Zero bytes of document or token data ever touch persistent secondary storage (NAND flash).

2.2 High-Level Product Functions

- **Multi-Factor Biometric Binding:** Simultaneous verification across facial biometrics, fingerprint recognition, and out-of-band primary passkey challenge.
- **Isolated Ephemeral Sandboxing:** Enforced separation of session runtime memory from host device OS subsystems.
- **Real-time Autonomous Fraud Dispatch:** Instant notification alerts to the user's registered hardware detailing merchant, GPS coordinates, device fingerprint, and authorization bounds.
- **Sub-second Cryptographic Shredding:** Immediate memory deallocation, cache zeroization, and cryptographic key wiping upon session termination.
- **Tiered Commercial Monetization:** Dynamic feature gating and risk entitlement managed via RevenueCat.

2.3 User Classes and Characteristics

[Architecture / Flow Diagram]

```
mindmap
  root((PhantomID Actors))
    Consumer User
      Free Tier
      Premium Tier
      Personal UPI & Emergency Access
    Enterprise User
      Corporate Provisioning
      SAML / OIDC Integration
      Client Terminal Access
    Host Device Owner
      Shared Circle / Friend
      Zero Session Visibility
      Host Runtime Host
    Enterprise Admin
      Audit Log Compliance
      Remote Session Revocation
      Trusted Device Fleet
```

1. **Consumer User:** Individuals seeking emergency liquidity, travel verification, or borrowing access. Interacts primarily with family, friends, or trusted circle nodes.
2. **Enterprise User:** Corporate employees, travelling auditors, and outsourced contractors who require isolated access to corporate assets on partner workstations.
3. **Host Device Owner:** The proprietor of the host hardware. Has zero cryptographic visibility into the session contents; can only inspect session status (Active / Terminated).
4. **Enterprise System Administrator:** Oversees corporate fleet policies, reviews compliance audits, and provisions device pools.

2.4 Operating Environment

- **Mobile Operating Systems:** Android 10.0+ (API Level 29+); iOS 14.0+.
- **Hardware Security Requirements:** Biometric authentication hardware (Class 3 Strong Biometric sensor for Android; Secure Enclave / FaceID / TouchID for iOS).
- **Backend Infrastructure:** Python 3.11+ (FastAPI) asynchronous microservices / Node.js 20+ LTS services.
- **Database & Ephemeral Cache:** Supabase (PostgreSQL 15+) with Row Level Security (RLS); Redis 7.x with sub-second TTL expiry keys.
- **Network Requirements:** Dual-ended HTTPS/WSS internet connectivity (min 4G / LTE or Wi-Fi).

2.5 Design & Implementation Constraints

[!WARNING] **Constraint Transparency:** 1. **OS Sandboxing Depth:** Complete kernel-level clipboard segregation and screen capture negation on unrooted consumer devices requires OEM-level OS privileges or MDM profiles. The hackathon MVP implements app-layer window flags (FLAG_SECURE , iOS view protection) and clearly documents this boundary. 2. **Financial Clearing Rail:** Real-time production UPI settlement requires sponsor bank certification and NPCI clearance. The MVP simulates execution using deep-link bridging and test sandbox rails.

3. Functional Requirements

3.1 FR-01: Registration & Onboarding

ID	Requirement Specification	Priority
FR-01.1	The system shall register consumer users via E.164 compliant phone numbers verified through high-entropy cryptographic OTP.	HIGH
FR-01.2	The system shall enroll at least one primary biometric factor (FIDO-aligned facial geometry or biometric fingerprint).	HIGH
FR-01.3	The system shall register a FIDO2 WebAuthn Passkey strictly bound to the hardware Secure Enclave of the primary device.	HIGH
FR-01.4	The system shall generate a cryptographically non-reversible User Identity Token (UIT) anchored to user identity.	HIGH
FR-01.5	The system shall support enrollment of a "Trusted Circle" comprising verified peer phone numbers.	MEDIUM

3.2 FR-02: Multi-Factor Authentication (MFA)

ID	Requirement Specification	Priority
FR-02.1	The host application shall capture and attest the initiator's live facial biometric vector.	HIGH
FR-02.2	The host application shall verify biometric fingerprint attestation via native hardware APIs.	HIGH
FR-02.3	The backend shall transmit an out-of-band cryptographic challenge payload to the initiator's primary passkey device.	HIGH
FR-02.4	The system shall mandate 100% convergence across all three verification factors prior to session generation; zero partial elevation permitted.	HIGH
FR-02.5	The backend shall enforce progressive rate-limiting, locking authentication attempts for 300 seconds following 3 sequential factor rejections.	HIGH
FR-02.6	The system shall support fallback time-based out-of-band SMS OTP verification during emergency recovery scenarios.	LOW

3.3 FR-03: Ephemeral Session Lifecycle

[Architecture / Flow Diagram]

stateDiagram-v2

[*] --> Dormant

Dormant --> MFA_Pending: Initiator Scans Host QR

MFA_Pending --> Active: 3-Factor Biometric & Passkey Validated

MFA_Pending --> Dormant: Factor Failed / Rate Limited

state Active {

[*] --> ExecutingTransaction

ExecutingTransaction --> AwaitingBiometricConfirm: Amount > Threshold

AwaitingBiometricConfirm --> ExecutingTransaction: Confirmed

}

Active --> Terminated: Payment Done / Manual Exit / Timeout / App Backgrounded

Terminated --> Purged: Overwrite & Zeroize Memory (<1s)

Purged --> [*]

ID	Requirement Specification	Priority
FR-03.1	The system shall instantiate an in-memory isolated execution container upon MFA confirmation.	HIGH
FR-03.2	The session initialization payload shall include: UUIDv4 Session ID, Ephemeral AES-256 GCM Token, Real-Time Device Trust Score, and Backend Signature.	HIGH
FR-03.3	Host runtime shall continually validate session token vitality against the backend edge-router via full-duplex WebSockets.	HIGH
FR-03.4	The session shall enforce a strict non-extendable inactivity timeout (default: 600 seconds).	HIGH
FR-03.5	The runtime shall trigger immediate lifecycle termination upon: transaction execution, user "Terminate" trigger, OS window blur, device locking, or app backgrounding.	HIGH
FR-03.6	On termination, the client runtime shall perform cryptographic zeroization: overwriting and releasing all heap buffers, cache partitions, and cryptographic keys within 1,000 milliseconds.	HIGH
FR-03.7	The backend shall archive an anonymized cryptographic audit digest excluding all PII data.	MEDIUM

3.4 FR-04: Session Permission Boundaries

3.4.1 Allowed Inside Session (Whitelisted Runtime)

- Initiation of verified UPI transaction payloads.
- Read-only account balance querying.
- Ephemeral inspection of unencrypted document views (rendered directly into memory).
- Camera scanning of external payment QR codes.
- Execution of signed cryptographic digital signatures.

3.4.2 Blocked by Design (Blacklisted Subsystems)

- Access to host operating system contacts, photo libraries, or media storage.
- System clipboard inspection or modification (clipboard access blocked via platform runtime wrappers).

- Screen capture, broadcast, and recording mechanisms (`FLAG_SECURE` on Android, screen obfuscation on iOS).
- Inter-process communication (IPC) with unverified local host processes.
- Network socket operations targeted to any IP/domain beyond PhantomID verified gateways.

3.5 FR-05: Financial Payment Execution

ID	Requirement Specification	Priority
FR-05.1	The session runtime shall generate NPCI-compliant UPI payment intent payloads.	HIGH
FR-05.2	The runtime shall interface with camera hardware for real-time BharatQR / UPI QR scanning.	HIGH
FR-05.3	High-value payments shall enforce step-up biometric re-verification prior to clearing dispatch.	HIGH
FR-05.4	The backend shall issue a cryptographically signed digital receipt containing transaction timestamp, randomized identifier, and hash digest.	HIGH
FR-05.5	The MVP shall implement UPI deep-link intent routing and mock settlement ledger simulation.	HIGH
FR-05.6	The runtime shall provide read-only balance verification without allowing balance transfer manipulation.	MEDIUM

3.6 FR-06: Fraud Detection & Anomaly Notification

ID	Requirement Specification	Priority
FR-06.1	The backend shall dispatch a high-priority push notification (FCM / APNs) to the primary hardware within 5,000ms of session instantiation.	HIGH
FR-06.2	The alert payload shall convey: Host Device Hardware Fingerprint, Approximate Geolocation, Merchant Counterparty, Amount, and Unix Timestamp.	HIGH
FR-06.3	The primary user app shall present dual action paths: "Recognized" or "Dispute & Terminate" .	HIGH
FR-06.4	A dispute invocation shall trigger immediate freeze protocols, cross-referencing biometric attestation telemetry and audit logs.	HIGH
FR-06.5	The system shall not process automated chargeback refunds upon dispute; an investigative quarantine pipeline shall be enforced.	HIGH
FR-06.6	An AI-driven risk model shall evaluate behavioral signals, flagging abnormal sessions prior to container generation.	MEDIUM

3.7 FR-07: Commercial Subscription Tiers (RevenueCat)

Feature Dimension	Free Tier	Premium Tier	Enterprise Tier
Monthly Ephemeral Sessions	5 sessions	Unlimited	Uncapped / Policy Governed
AI Dynamic Risk Scoring	Disabled	Real-Time Evaluation	Advanced Threat Modeling
Travel Mode / Roaming Bounds	Disabled	Enabled	Enabled
Trusted Family Federation	Disabled	Up to 5 Members	Multi-Tenant Org Control
Encrypted Audit Log Cloud Sync	Disabled	30-Day Retention	Infinite SIEM Export
SLA & Support Escalation	Community	Priority 24/7	Dedicated Account Engineer
Custom Trusted Device Pools	Disabled	Disabled	Configurable Fleet Whitelist
MDM / EMM Infrastructure Link	Disabled	Disabled	Zero-Trust MDM Integration

4. Non-Functional Requirements

4.1 Security Requirements (NFR-SEC)

- **NFR-SEC-01:** Session tokens and transit payloads must be encrypted using AES-256-GCM with unique initialization vectors (IVs).
- **NFR-SEC-02:** All transport layer communications must strictly enforce TLS 1.3 with SHA-256 certificate pinning.
- **NFR-SEC-03:** Raw biometric minutiae, facial embeddings, and private keys must never leave local device secure enclaves.
- **NFR-SEC-04:** Cryptographic keys must be strictly ephemeral; session keys must be generated using cryptographically secure random number generators (CSPRNG) and discarded post-use.
- **NFR-SEC-05:** Server-side audit archives must be pseudonymized, stripped of PII, and encrypted at rest utilizing AWS KMS / Vault HSM keys.
- **NFR-SEC-06:** Client mobile binaries must incorporate root/jailbreak detection, debugger attachment prevention, and binary repackaging tamper checks.
- **NFR-SEC-07:** The software design must conform strictly with OWASP Mobile Security Testing Guide (MSTG Level 2).

4.2 Performance Requirements (NFR-PERF)

- **NFR-PERF-01:** Ephemeral session generation latency post-MFA must not exceed 3,000 milliseconds over standard 4G/LTE cellular connections.
- **NFR-PERF-02:** Complete cryptographic memory zeroization and container termination must complete within 1,000 milliseconds.
- **NFR-PERF-03:** Anomaly fraud push notifications must arrive on the primary device within 5,000 milliseconds of session instantiation.
- **NFR-PERF-04:** Total multi-factor biometric attestation pipeline latency must complete in under 5,000 milliseconds.
- **NFR-PERF-05:** Cloud API gateway architecture must support 10,000 concurrent sessions (MVP demo scale target: 100 concurrent streams).

4.3 Reliability & Fault Tolerance (NFR-REL)

- **NFR-REL-01:** Core API infrastructure must guarantee 99.9% uptime availability.
- **NFR-REL-02:** If client network disconnects mid-session, the client runtime must enter immediate failsafe termination and wipe all session data locally.
- **NFR-REL-03:** All financial API mutations must incorporate idempotency keys (`Idempotency-Key: UUIDv4`) to eliminate duplicate debit hazards.

4.4 Usability & Human Interface (NFR-UX)

- **NFR-UX-01:** Total elapsed time from host app opening to active transaction readiness must not exceed 30 seconds for first-time flows.
- **NFR-UX-02:** Session countdown timer, trust status indicators, and prominent single-touch "Kill Session" buttons must remain fixed in the primary viewport.
- **NFR-UX-03:** Full localization support for English and Hindi in the MVP release.

4.5 Privacy & Regulatory Compliance (NFR-PRIV)

- **NFR-PRIV-01:** Architectural design must comply with the provisions of the India Digital Personal Data Protection Act (DPDPA), 2023.
 - **NFR-PRIV-02:** Host device owners must be given zero telemetry or access regarding the initiator's transactions, document contents, or account balances.
 - **NFR-PRIV-03:** Users must be empowered with an automated "Right to Erasure" interface allowing irreversible purging of cloud session metadata.
-

5. System Architecture & Engineering Specifications

5.1 Three-Tier Architecture

[Architecture / Flow Diagram]

```
graph TB
    subgraph Client_Tier [Client Tier]
        HostApp["Host Device App (React Native)"]
        UserApp["Primary Device App (Passkey / FCM)"]
    end

    subgraph Edge_API_Tier [Edge & API Tier]
        Gateway["Cloudflare API Gateway & WAF"]
        FastAPI["Backend Engine (FastAPI / Node.js)"]
        Redis["Ephemeral Redis Cache (TTL Key Eviction)"]
    end

    subgraph Persistence_Services [Persistence & Services]
        Supabase["Supabase PostgreSQL + RLS"]
        OpenAI["OpenAI Risk Scoring Engine"]
        RC["RevenueCat Billing Infrastructure"]
        FCM["Firebase Cloud Messaging (FCM)"]
    end

    HostApp <-->|TLS 1.3 / WSS| Gateway
    UserApp <-->|TLS 1.3 / Push| Gateway
    Gateway --> FastAPI
    FastAPI <--> Redis
    FastAPI --> Supabase
    FastAPI --> OpenAI
    FastAPI --> RC
    FastAPI --> FCM
```

5.2 Core Module Responsibility Matrix

Subsystem Module	Primary Responsibility	Technology Implementation
Authentication Engine	Multi-factor verification; orchestration of face, fingerprint, and WebAuthn challenges.	WebAuthn, Android BiometricPrompt, iOS LocalAuthentication
Session Manager	Lifecycle state machine; issuance of ephemeral session tokens; memory eviction.	FastAPI Async Workers, Redis ephemeral store
AI Risk Engine	Dynamic risk calculation evaluating device anomaly, geolocation variance, and history.	OpenAI API (MVP Demo Model), Isolation Forests (Production)
Payment Facilitator	Construction of signed UPI payment intent envelopes and BharatQR decoding.	Deep-link bridge, NPCI Sandbox simulator
Entitlement Service	Tier validation, active feature flags, subscription lifecycle management.	RevenueCat Mobile SDK
Compliance Audit Ledger	Immutable recording salted, pseudonymized session hashes for post-incident analysis.	PostgreSQL Row Level Security (RLS)

5.3 Sequence Flow: Complete Session Lifecycle

[Architecture / Flow Diagram]

```
sequenceDiagram
    autonumber
    actor Initiator as Initiator User
    participant Host as Host Device (PhantomID)
    participant Cloud as PhantomID Cloud Gateway
    participant Primary as Primary Phone (Passkey)
    participant Bank as UPI / Payment Rail

    Initiator->>Host: Open PhantomID App & Initiate Session
    Host->>Cloud: Request New Session Challenge
    Cloud-->>Host: Issue Nonce & Challenge Parameters
    Host->>Initiator: Prompt Facial & Fingerprint Biometric
    Initiator->>Host: Provide Biometrics (Hardware Attested)
    Cloud->>Primary: Push WebAuthn Passkey Challenge
    Primary->>Initiator: Prompt Passkey Confirmation
    Initiator->>Primary: Confirm via Primary Secure Enclave
    Primary-->>Cloud: Return Cryptographic Attestation
    Cloud->>Cloud: Verify 3 Factors & Calculate Risk Score
    Cloud-->>Host: Inject Ephemeral AES-256 Session Capsule
    Cloud->>Primary: Send Real-Time Alert Notification
    Note over Host: Sandboxed Session Active
    Initiator->>Host: Scan QR & Authorize ₹500 Payment
    Host->>Bank: Dispatch Signed UPI Intent Payload
    Bank-->>Host: Transaction Success Receipt
    Host->>Host: Trigger Automated Shredding Routine
    Note over Host: Zeroize RAM, Wipe Cache & Invalidate Token
    Host-->>Initiator: Display Ephemeral Success & Terminate
    Host->>Cloud: Notify Session Purged
```

6. Technology Stack (Hackathon MVP)

Architecture Layer	Component Selection	Engineering Rationale
Frontend Framework	React Native (Expo / Bare Workflow)	Rapid cross-platform native compilation for iOS and Android.
Backend Framework	FastAPI (Python 3.11)	High-concurrency async I/O, auto-generated OpenAPI, native ML compatibility.
Primary Database	Supabase (PostgreSQL 15)	Instant edge APIs, built-in row-level security (RLS), real-time subscriptions.
Ephemeral Cache	Redis Cloud	Sub-millisecond read/write latency with auto-expiring TTL keys for session tokens.
Authentication Standard	WebAuthn / FIDO2 + Platform Biometrics	Hardware-backed public key cryptography; zero stored passwords.
Risk Scoring AI	OpenAI API (GPT-4o Mini)	Low-latency behavioral risk classification demonstration during hackathon judging.
Subscription Gateway	RevenueCat SDK	Cross-platform purchase receipts, dynamic feature tier entitlements.
Push Notification Rail	Google Firebase Cloud Messaging (FCM)	Reliable low-latency out-of-band delivery to primary hardware.

6.1 Scope Boundary: Hackathon MVP vs. Production

[Architecture / Flow Diagram]

```
classDiagram
    class Hackathon_MVP {
        +App-level window flags (FLAG_SECURE)
        +UPI Deep-link intent & mock ledger
        +OpenAI API prompt-based risk score
        +Platform LocalAuthentication APIs
        +Standard Supabase audit logging
    }
    class Commercial_Production {
        +OEM & Enterprise MDM Sandbox Profiles
        +Certified NPCI Payment Aggregator Bridge
        +Edge TensorRT / CoreML behavioral models
        +Hardware FIDO2 Security Key & NFC Tokens
        +Tamper-evident Blockchain/SIEM Ledger
    }
    Hackathon_MVP <|-- Commercial_Production : Future Roadmap Evolution
```

7. Known Constraints, Risks & Assumptions

7.1 Technical Constraints

- 1. **Operating System Boundary:** Native mobile applications cannot prevent hardware display photography (analog hole) or root-level memory dump without custom enterprise MDM profiles.

2. **Payment Partner Clearance:** Direct NPCI switch access requires capital reserves, banking sponsorship, and CERT-In audits; the MVP relies on deep-link intent routing.

7.2 Business & Security Assumptions

- 1. The Host device possesses an uncompromised operating system baseline and active internet connectivity.
- 2. The Initiator maintains possession of their registered primary device to validate out-of-band Passkey challenges.
- 3. For enterprise contexts, client organizations enforce MDM compliance across their designated workstation pool.

8. Future Roadmap & Strategic Integrations

```
[Architecture / Flow Diagram]
timeline
    title PhantomID Strategic Evolution
    Phase 1 (Hackathon MVP) : Core 3-Factor Auth : Ephemeral Session Destruction : RevenueCat Tier Gating : UPI
Mock Flow
    Phase 2 (Consumer Beta Q1 2027) : Live NPCI UPI Clearing : DigiLocker Direct Integration : Trusted Circle
P2P
    Phase 3 (Enterprise Q3 2027) : MDM / EMM Profiles : Corporate SAML / OIDC SSO : Automated SIEM Export
    Phase 4 (Expansion 2028) : ABHA Health Dossier Access : Airline & Hotel Kiosk SDK : Aadhaar eKYC Gateway
    Phase 5 (Ecosystem 2029) : Sovereign W3C DID Credentials : Global Cross-Border Identity Capsules
```

9. Appendix

Appendix A: Requirement Priority Classification

- **HIGH (P0):** Mandatory for Hackathon MVP demonstration. System cannot function without this capability.
- **MEDIUM (P1):** Important architectural feature. Enriches product security and evaluation criteria.
- **LOW (P2):** Strategic long-term feature scheduled for post-MVP commercial production.

Appendix B: Document Revision History

Version	Publication Date	Primary Author	Summary of Modification
v1.0	11 September 2026	Himanshu Badgujar	Initial authoritative SRS publication based on PhantomID Pitch v1.0.

Appendix C: Author & Academic Affiliation

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